Bases de datos

NEREA MONTOYA OLIVARES

EXERCISE 1

Create the conceptual design of a database using the ER model that satisfies the following requirements:

a) We want to design a database to manage a company dedicated to the international transport of goods that operates throughout Europe.

b) The company has several delegations spread throughout the European geography. Delegations are identified by name, and they also want to register their telephone number. In a city there is never more than one delegation. We want to know the city where each delegation is located. It must be assumed that there are no cities with the repeated name (at least in the scope of this database). We also want to know the total area of each city.

c) Company staff can be separated into two large groups:

- Administrative, about which it is interesting to know their level of studies.

- Drivers, for whom it is interesting to know the year in which they obtained their driving license and the type of license they have.

About the company staff, we want to know the employee code (which identifies them), their name, their telephone number and the year of birth. All employees are assigned to a delegation. We want to have historical evidence of this fact, taking into account that they can change delegations (they can even return to a delegation where they had previously been). We want to know from which date they are in each delegation.

d) The activity of the company consists of making the relevant trips to transport the goods according to the requests of its clients. All customers are identified by a customer code. You also want to know the name and contact telephone number of each of them.

e) The company, to carry out its activity, has many trucks identified by a truck code. We need to know the license plate, make and tare of the trucks.

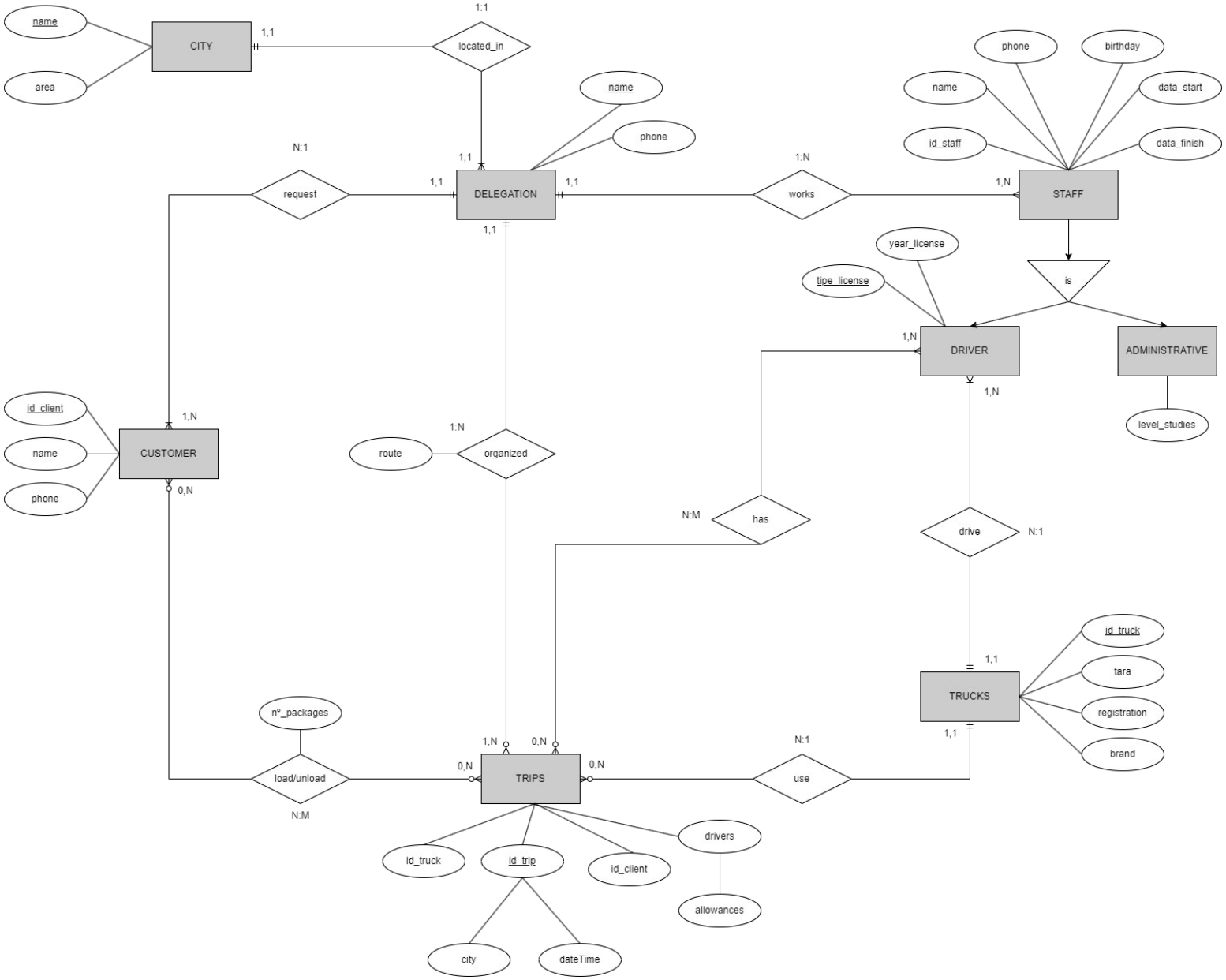
f) Trips are always organized by a delegation, and are identified by a trip code, which is internal to each delegation (and can be repeated in different delegations). For each of the trips that have been made, it is necessary to know:

- Which truck has been used (since each trip is made with only one truck).

- Which driver or drivers have gone (considering that several drivers can go on long trips). You also want to know the amount of allowances paid to each driver (taking into account that allowances may be different for different drivers on the same trip).

- The route of the trip; that is, the date and time the truck arrives at each of the cities where it must load or unload. We will assume that a trip never passes through the same city twice.

- The number of packages loaded and packages unloaded in each city, and for each of the clients. On the same trip, packages can be dropped off and/or picked up in different cities at the request of the same client. Also, on the same trip, packages can be dropped off and/or picked up in the same city at the request of different clients.


name
area
CITY
1,1
1:1
located_in
name
phone
DELEGATION
N:1
request
1,1
1,1
1,1
1:N
works
1,N
STAFF
phone
birthday
name
data_start
id_staff
data_finish
year_license
tipe_license
is
1,N
DRIVER
ADMINISTRATIVE
level_studies
1,N
id_client
name
phone
CUSTOMER
1,N
0,N
route
1:N
organized
N:M
has
drive
N:1
id_truck
tara
registration
brand
1,1
TRUCKS
1,1
n°_packages
load/unload
N:M
1,N
0,N
0,N
TRIPS
0,N
N:1
use
id_truck
id_trip
id_client
drivers
allowances
city
dateTime

EXERCISE 2

It is necessary to design a database for a football club manager to control interesting data about competitions, clubs, coaches, ... of state scope.

● Every season the clubs compete in several competitions (league, cup...) against each other. Our manager wants historical information on the classifications got by the clubs in the different competitions throughout all the seasons. The ranking will be specified by a position number: 1 means champion, 2 means runner-up, etc. Both competitions and seasons will have a code that identifies them, a name and a description.

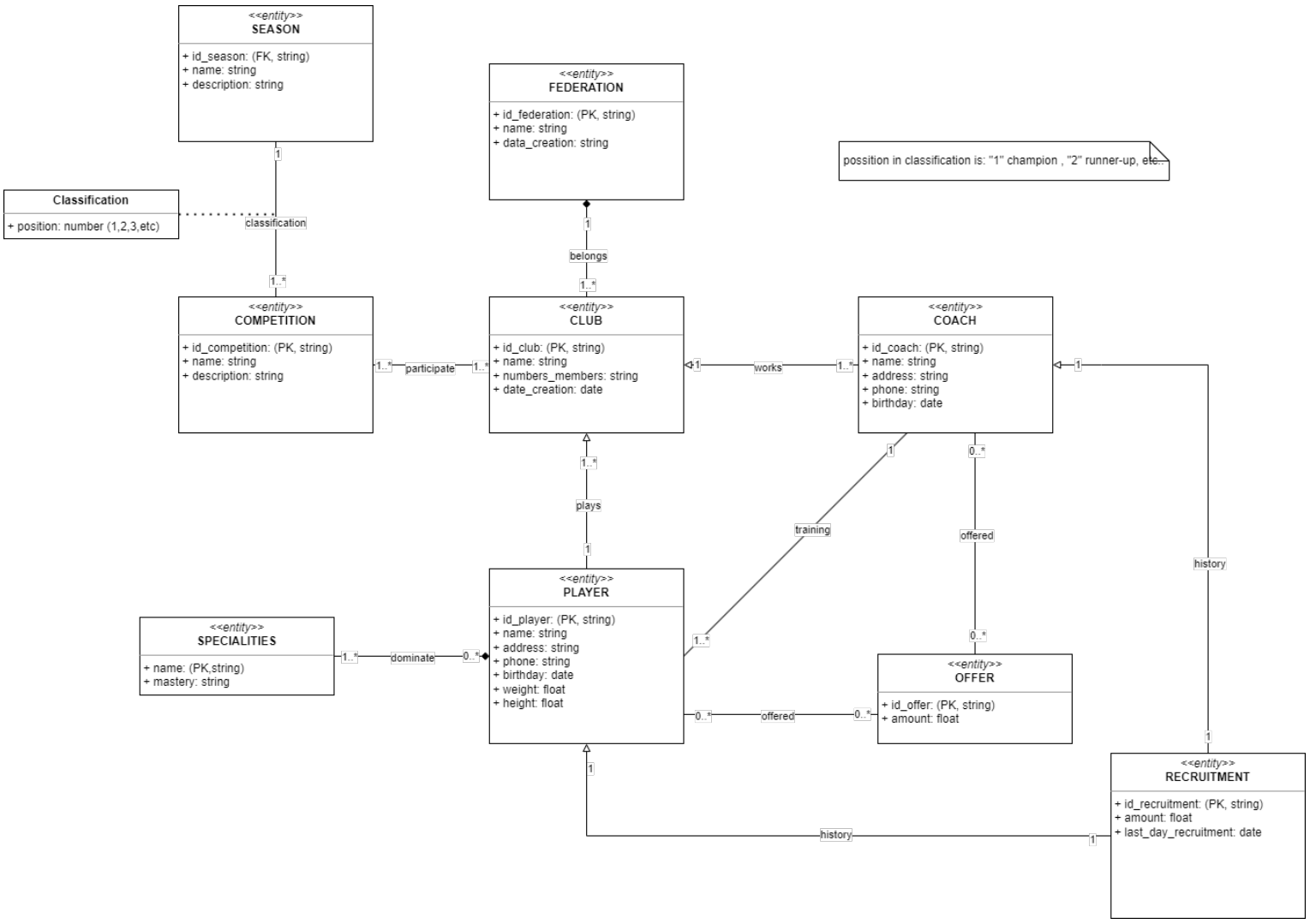
● The different clubs are grouped in the corresponding regional federations. Every federation has at least one club. He wants to know the name and date of creation of the federations, and of the clubs, the name and number of members.

● Information about players and coaches is very important. They will be identified by a code, and he wants to know everyone's name, address, phone number and date of birth. It should be mentioned that some coaches may have been players in their youth. He also wants to know the players' weight, height, specialty or specialties and what mastery they have (degree of specialty). Every player must have at least one specialty, but there may be specialties that no player has. Of the coaches he is
interested in the date on which they started their career as football coaches.

● Of all the people listed in the database (players and coaches), you want to know the history of recruitment by the various clubs, including the amount and the last day of each recruitment. A person may be employed by a single club at a given time, but may change clubs later, and may even return to a club they have previously worked for.

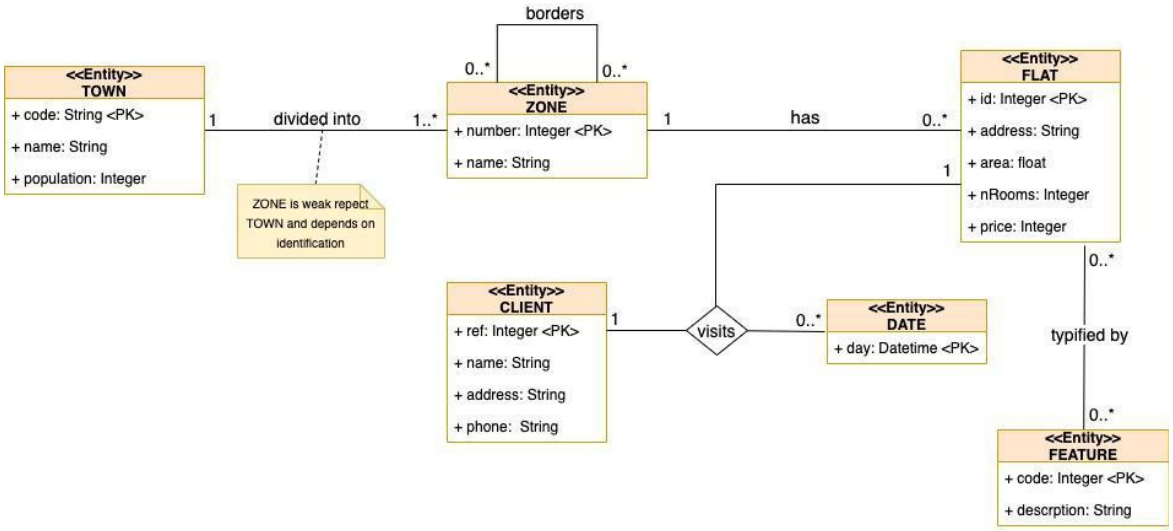
● It also wants to record the offers that the people in the database have received from clubs during their sporting life. It's important to have a record of the amount of the offers. It must be taken into account that at a given moment a person can receive many offers as long as they come from different clubs.

Do the conceptual design, using a UML class diagram, and collect all the semantics of the statement. You must indicate the entity types attributes with its corresponding data types, the relationships cardinalities, all the restrictions (primary keys, alternatives ...), as well as the requirements that have not been reflected in the schema proposed. If any
additional semantic assumptions have been made, it must also be reflected on the document with detailed arguments.


<<entity>>
SEASON
+ id_season: (FK, string)
+ name: string
+ description: string
<<entity>>
FEDERATION
+ id_federation: (PK, string)
+ name: string
+ data_creation: string
possition in classification is: "1" champion , "2" runner-up, etc...
Classification
+ position: number (1,2,3,etc)
classification
belongs
<<entity>>
COMPETITION
+ id_competition: (PK, string)
+ name: string
+ description: string
participate
<<entity>>
CLUB
+ id_club: (PK, string)
+ name: string
+ numbers_members: string
+ date_creation: date
works
<<entity>>
COACH
+ id_coach: (PK, string)
+ name: string
+ address: string
+ phone: string
+ birthday: date
plays
training
offered
history
<<entity>>
SPECIALITIES
+ name: (PK,string)
+ mastery: string
dominate
<<entity>>
PLAYER
+ id_player: (PK, string)
+ name: string
+ address: string
+ phone: string
+ birthday: date
+ weight: float
+ height: float
<<entity>>
OFFER
+ id_offer: (PK, string)
+ amount: float
offered
<<entity>>
RECRUITMENT
+ id_recruitment: (PK, string)
+ amount: float
+ last_day_recruitment: date
history

EXERCISE 3

Given the following conceptual model:



Transform to the Relational model and detail the tables, attributes, keys and foreign keys generated, indicating whether, due to the transformation, an attribute can take a NULL value.

Town (townCode <PK>, name, population)

Zone (numberZone <PK>, name, townCode <FK>)

BorderZone (border <PK>, numberZone <FK>)

Flat (idFlat <PK>, address, area, n_rooms, price, numberZone <FK>)

Typified (idFlat <PK>, idCode <PK>)
*This is a new table created consisting of the two <PKs> of flat and features

Feature (featureCode <PK>, description)

Date (idDay <PK>)

Client (refClient <PK>, name, address, phone)

Visits (refClient <PK>, idDay <PK>, idFlat <FK>)
*This is a new table created consisting of the two <PKs> of client and date and one <FK> for flat.

EXERCISE 4

We want to design a relational database to store information about the matters that a law firm handles. Each matter has a file number that identifies it and corresponds to a single client. The start date of the matter must be stored, the end date, its status (in process, archived, etc.), as well as the personal data of the client to which it belongs (ID, name, address and phone). Obviously, a client can have more than one matter. Some matters are handled by one or more lawyers and vice versa, whose personal data (ID, name and phone) is also of interest to us. We also need to know the first day that each lawyer starts to work on a matter.

Imagine that, as part of the corresponding relational model, we have the following relation (attributes that form the primary key are underlined and in bold):

LawyerFirm (ID_client, name_client, address_client, phone_client,**nr_matter**, start_date, end_date, status, **ID_lawyer**, name_lawyer, phone_lawyer, firts_date)

1. Indicate if the LawyerFirm is in FNBC. Explain your answer in detail.

Not in FNBC model because there are multivalued attributes.

2. If it is not in FNBC, describe what is the problem and make an equivalent relational model standardized to FNBC. Point out the primary and external keys to the relationships obtained.

The problem is when attributes depend on candidate primary keys and not only on the primary key.

All the relations are already in FNBC:

Client (ID_client <PK>, name_client, address_client, phone_client)

*The Client table stores information about Clients.

Lawyer (ID_lawyer <PK>, name_lawyer, phone_lawyer, firts_date,)

*The Lawyer table stores information about Lawyer.

Matter (nr_matter <PK>, start_date, end_date, status, ID_client<FK>)

*The Matter table stores information about Matter.

LawyerMatter (nr_matter <PK>, ID_lawyer <PK>)

*The LawyerMatter table has two <PK> (nr_matter and id_lawyer) so we get the information from two tables in one.